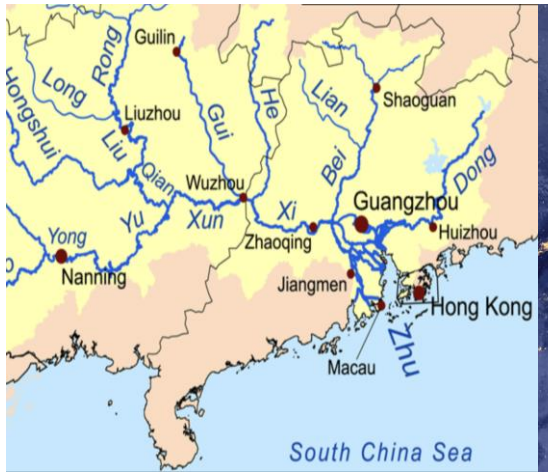
2026 and Beyond for IT

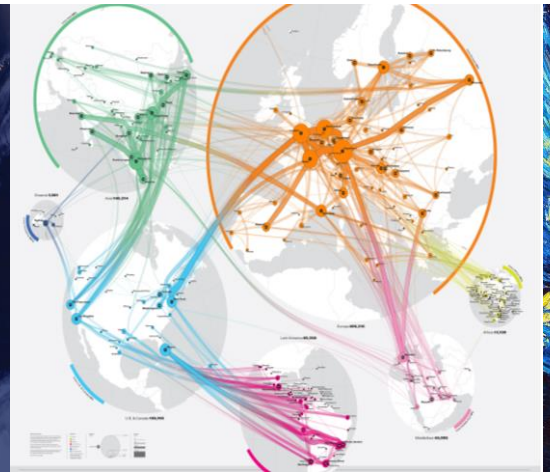
What’s the Difference?

Back to Fundamentals



Water

Electricity



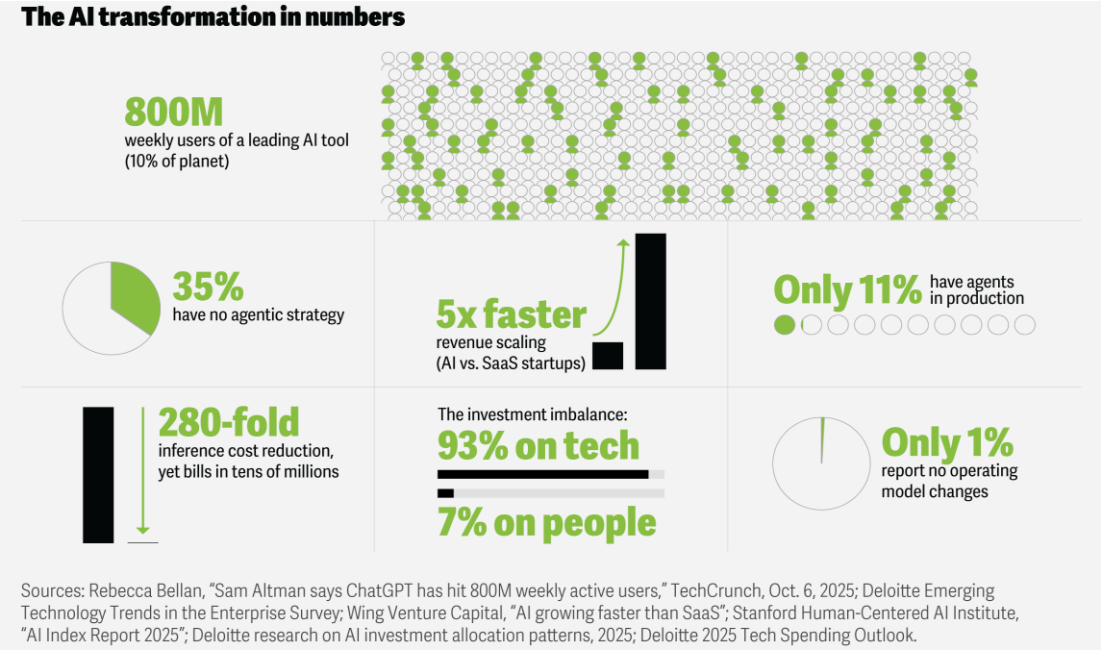
Internet

Token

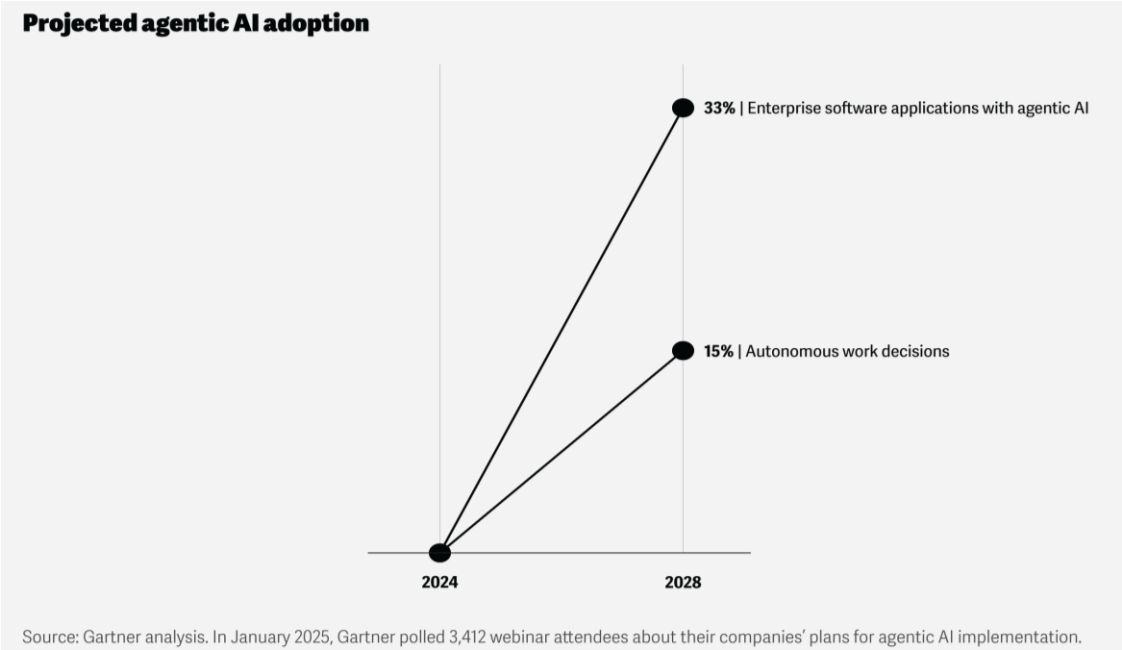
AI - The Backbone of Enterprise IT

AI is no longer a nice-to-have feature — it is becoming the core infrastructure of IT (Gartner, Deloitt)

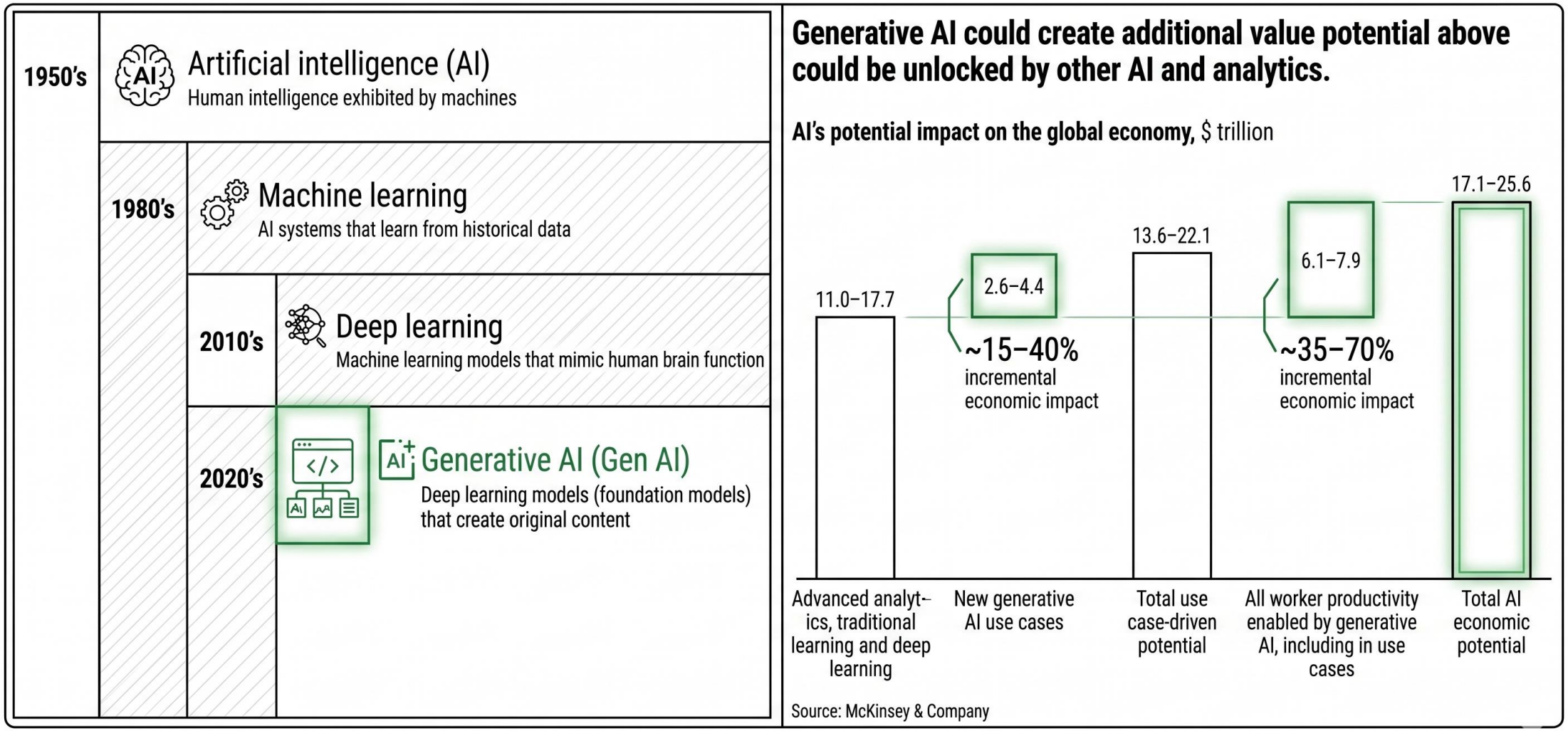
Innovation in Machine Speed



Silicon-based Workforce



AI Evolution



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The Five-Layer AI Infrastructure

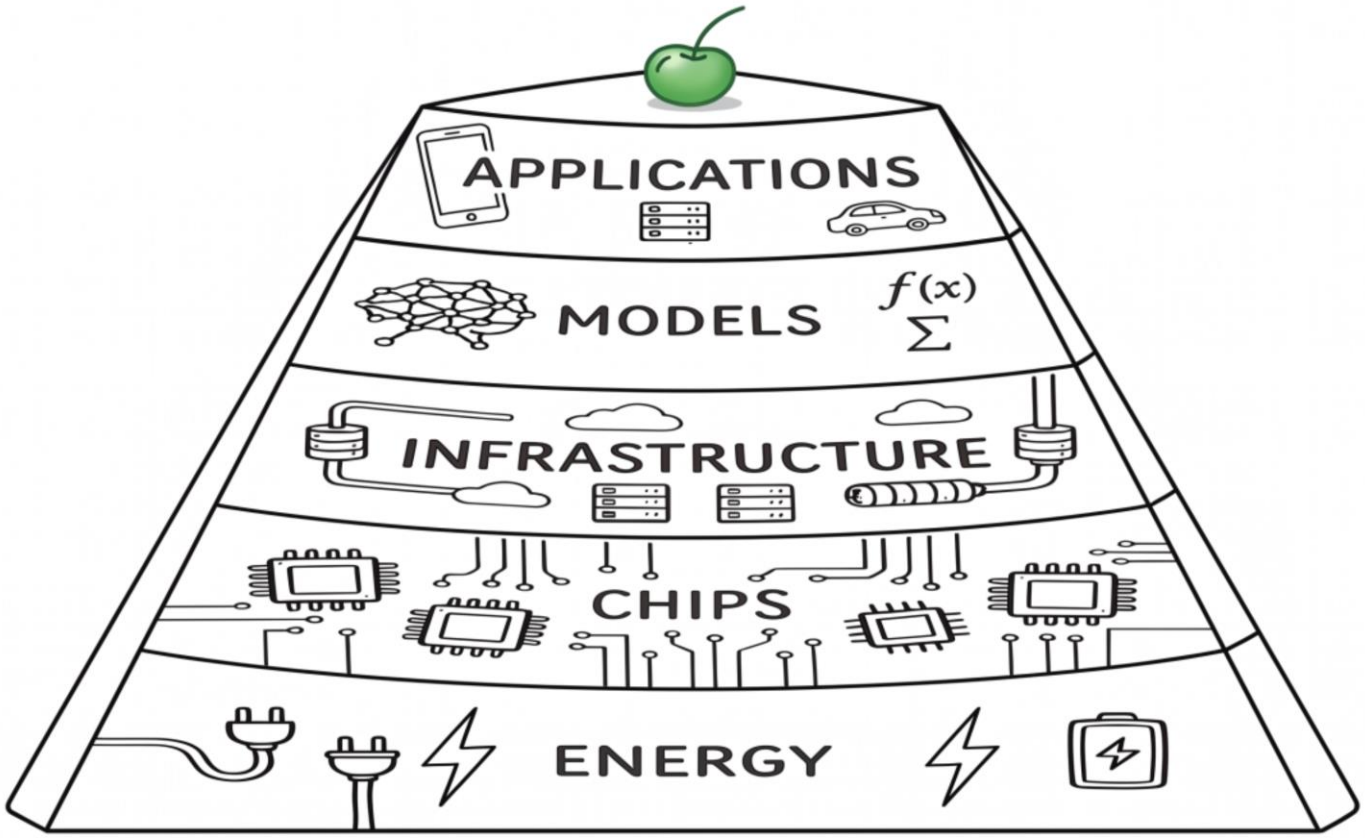
Layer 5: Applications (Claws?)
The end-user products and services that leverage AI to solve problems and provide value.

Layer 4: models
The software algorithms and neural networks (including open-source) that contain and form the intelligence of AI.

Layer 3: Infrastructure
The physical data centers and data infrastructure that house, cool, support and operate the intelligence and its related workloads.

Layer 2: Chips
The specialized semiconductors and processors that perform the complex calculations for AI tasks.

Layer 1: Energy
The foundational power required to run all AI systems, supercomputers, and data centers.



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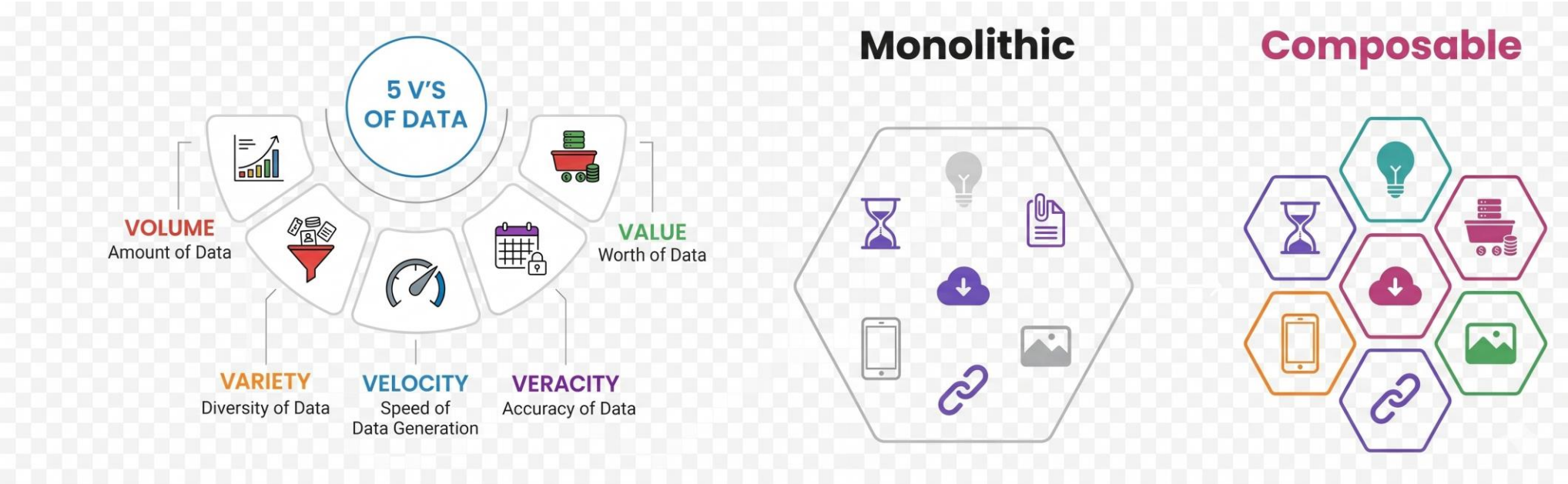
Strategic Hybrid for Sovereign AI

- *Unpredictable Workload, Non-sustainable Advantage, Unmanageable Cost, Inconsistent Work Quality, Privacy and Security Challenging, …*
- Building IT strategy for Agentic AI and Inference economics
  - Token cost have dropped 280-fold in two years, yet some enterprises are seeing monthly bills in the tens of millions
  - The existing infrastructure strategies are not designed to scale AI to production-scale deployment
  - Shifting to strategic hybrid: on-premises for consistency & competence & sovereignty … and edge for immediacy, cloud for elasticity and scale with **a single unified data platform**
  - Multi-Agent is **the new composable platform**



Physical Infrastructure Impact – Token Economy : Data-Centric

AI infrastructure's success is measured by how fast STORAGE can feed GPUs with relevant DATA.

The most important metrics in the AI era are shifting from traditional KPIs to **AI pipeline metrics**:

Traditional Storage Metrics

- Capacity
- IOPS
- Reliability

AI-Pipeline Metrics

- **GB/s per GPU**
- **TTFT (Time to First Token)**
- **TPS (Tokens Per Second)**
- **PB-scale datasets**
- **TB/s cluster bandwidth**
- **Billions of vectors**
- **Data ingestion rate**
- **Global context and memory**

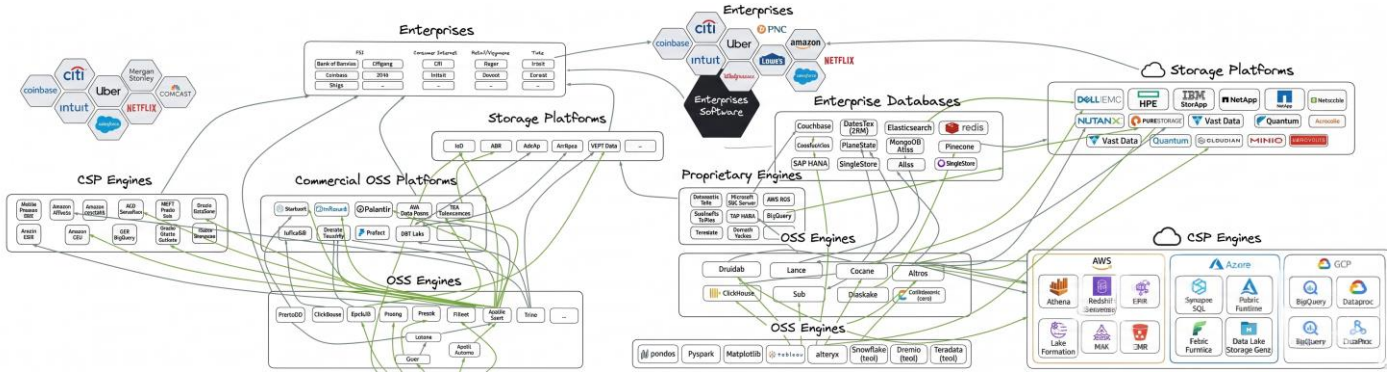
Infrastructure impact

Network upgrades:

- Ultra high-speed
- Fiber optical networks
- RDMA

Storage upgrades:

- Inference Context Memory Storage
- DPU Capability
- Global Namespace
- KV Cache Acceleration
- High-speed Data Path
- Massive Scalability



Enterprise Data Platforms – the Foundation for Intelligence

A whole new form of data platform is emerging for Structured Data and Unstructured Data

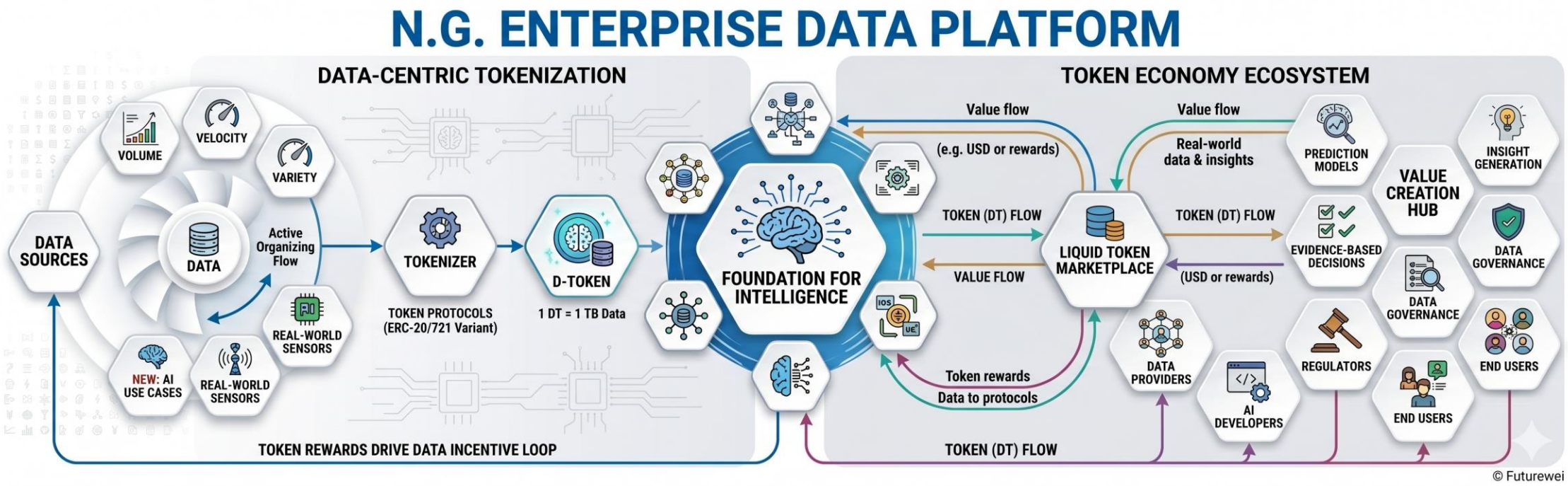
Application Impact

All applications require data to be:

- Discoverable
- Persistent
- Searchable

Infrastructure impact - Enterprises must store:

- context memory
- training datasets
- embeddings
- real-time telemetry
- prompts
- Logs
- … …



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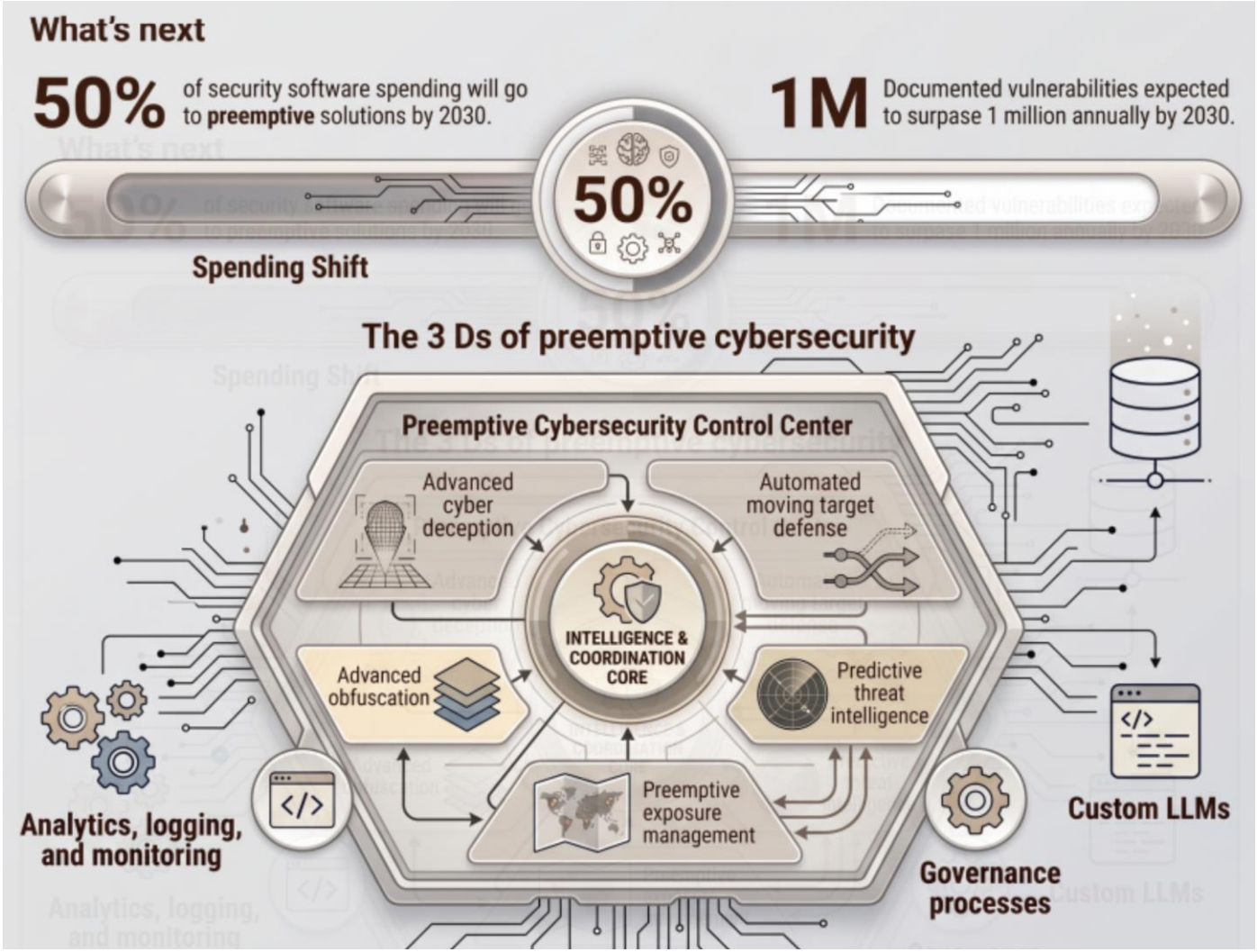
Preemptive Cybersecurity

The 3 D's of Preemptive Cybersecurity: Deny, Deceive and Disrupt (Gartner)



Storage #1: Inference Context Memory

AI Inference context Memory → Brand new storage demand

Forbes
https://www.forbes.com › Innovation › Enterprise Tech

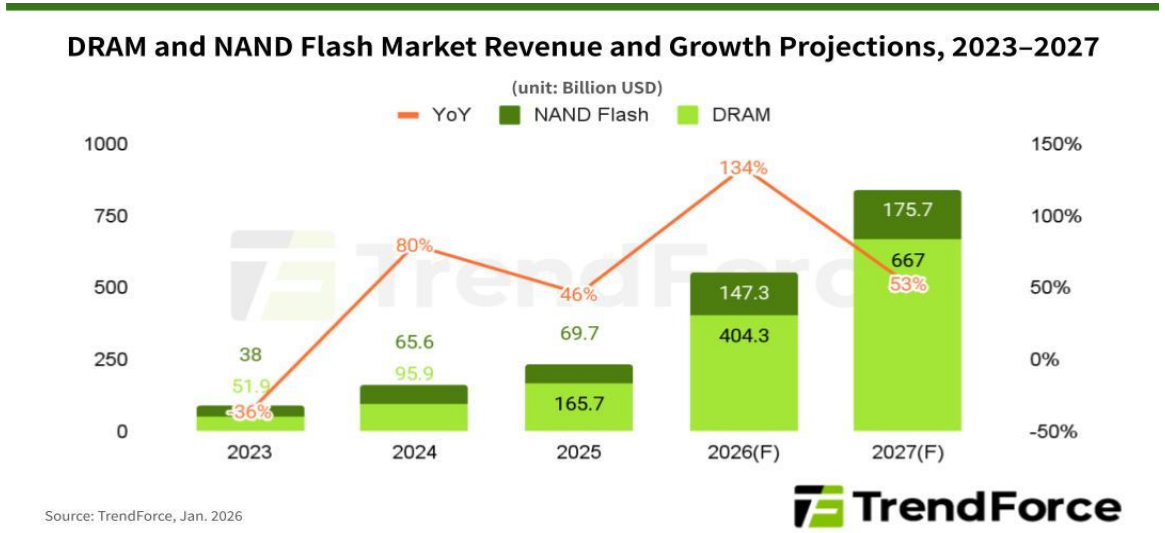
ICMSP Could Drive Additional 100EB Of AI Storage

16 Jan 2026 — The Nvidia ICMSP will store **16TB of storage per GPU** and this can enable petabytes of shared context across a GPU cluster to support very large ... Read more

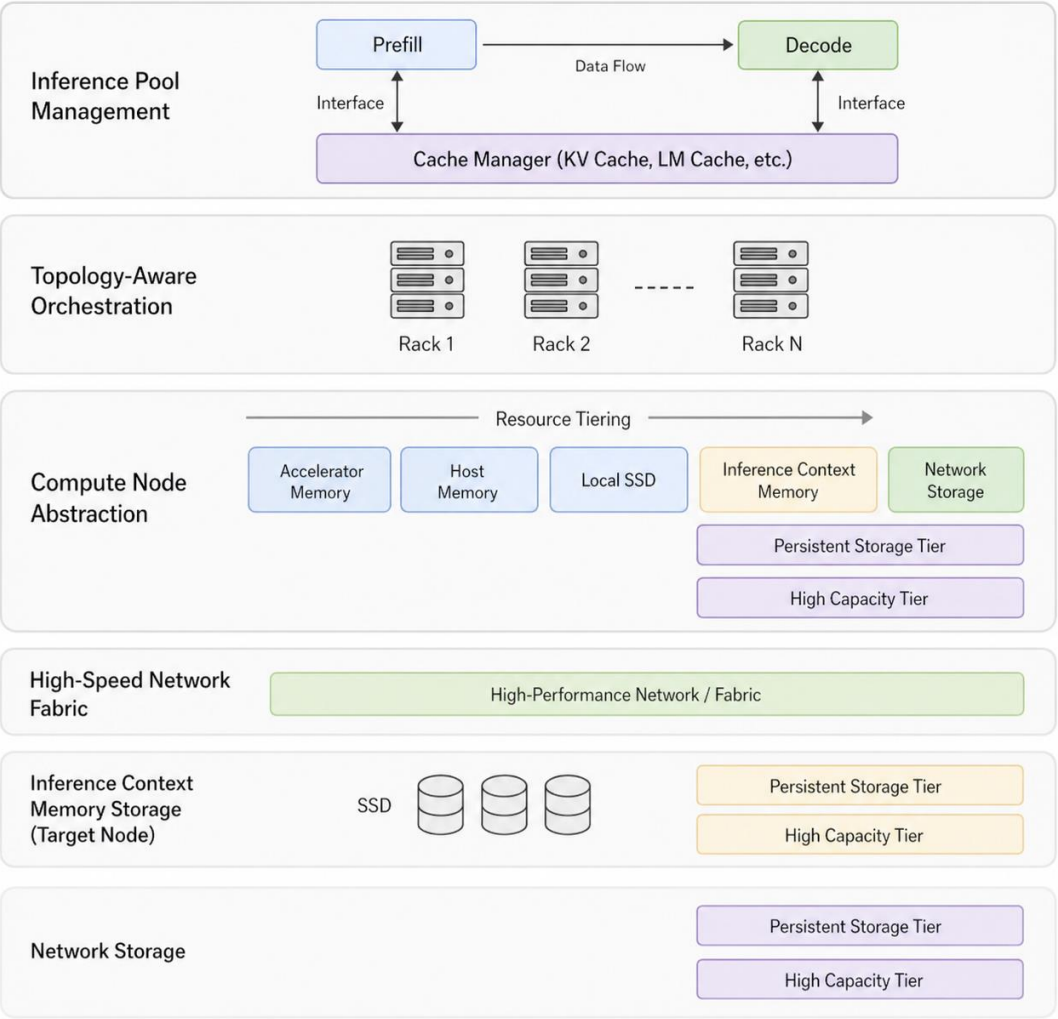
Third Bridge
https://www.thirdbridge.com › media › perspectives › e...

Why storage and memory are the new "AI Database" for AGI

19 Jan 2026 — Third Bridge spoke with industry experts to explore **why storage and memory are the new "AI Database" for AGI**, and the implications of Nvidia's ... Read more



ISMSP (Inference Context Memory Storage Platform)

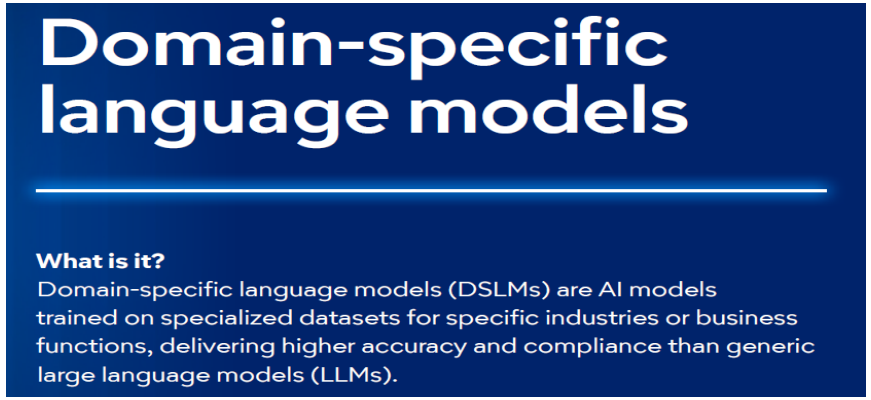


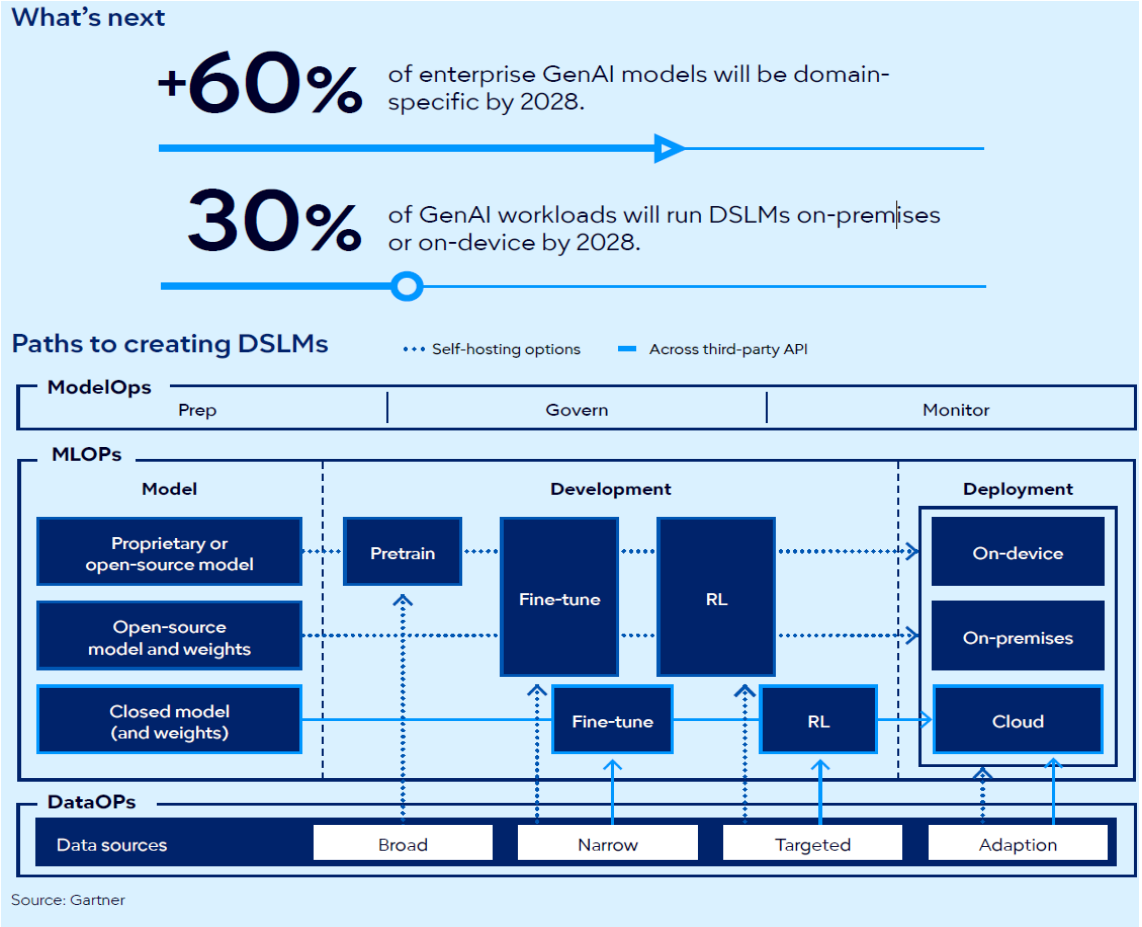
Storage #2: Data Explosion

AI is fundamentally a data problem → Storage demand explodes

Storage Demand

- Domain-specific language models (Gartner)
- AI training datasets – (PB-scale)
- Feature stores
- Vector knowledge base and RAG
- Model checkpoints





Storage #3: AI Factory

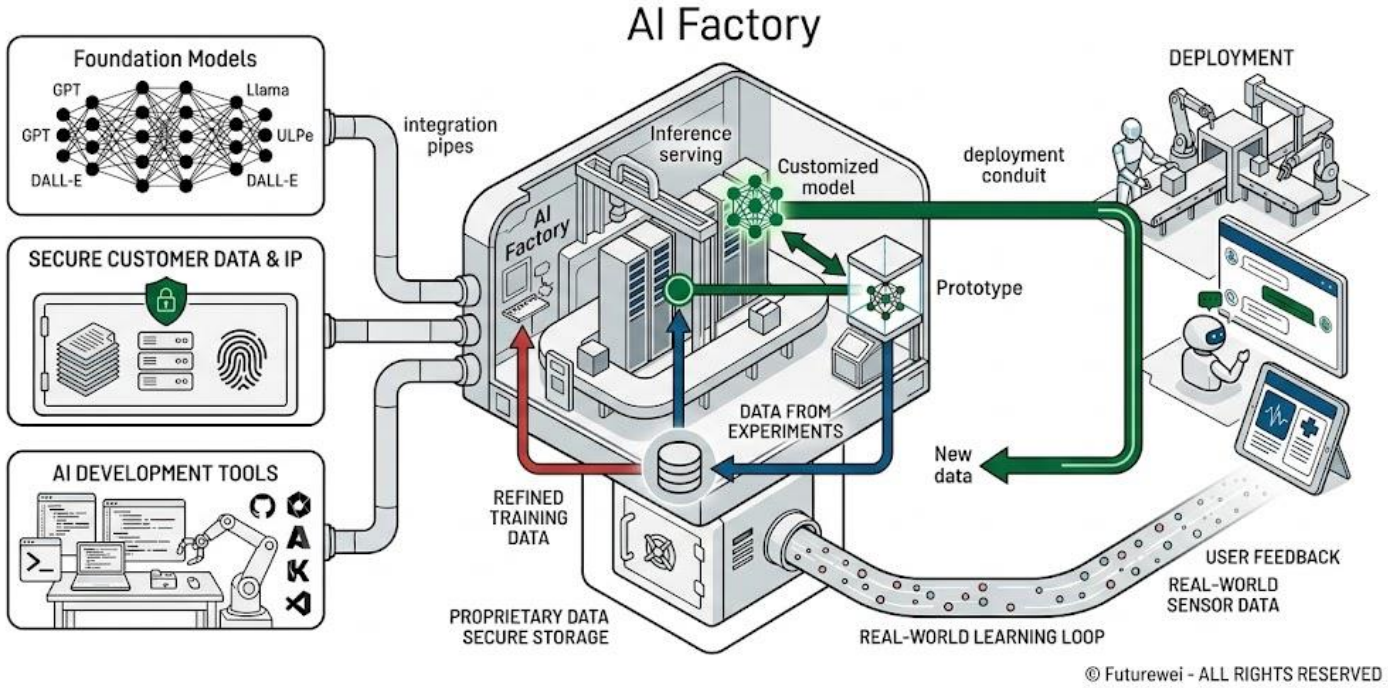
AI factory & AI clusters → Fast storage pipelines

Infrastructure impact
Companies are building:

- GPU clusters
- AI supercomputers
- Hyperscale AI data centers

Storage Requirements
These environments require:

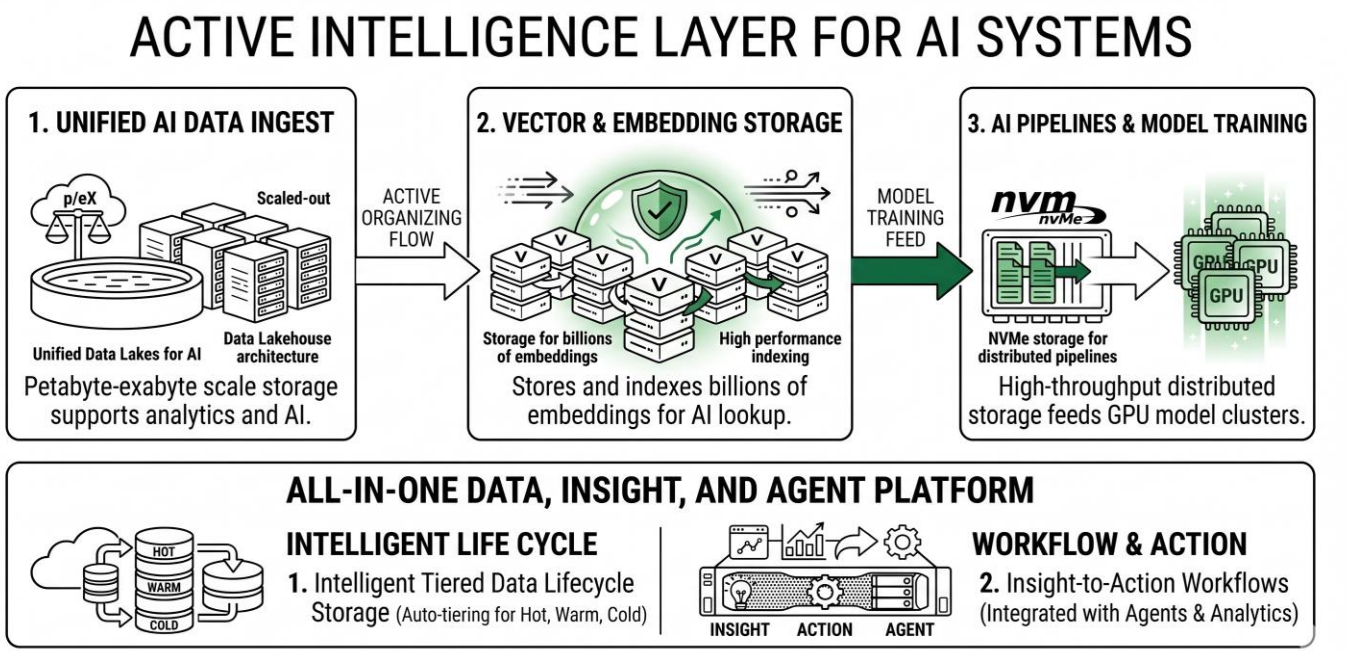
- High-performance storage
- Disaggregated architecture
- Distributed storage systems
- AI data lakes



Storage #4: AI Data Platform

Storage transforms from a **passive repository** into an **active intelligence layer** that continuously feeds, organizes, and enriches data for AI systems.

- All-in-one Data, Insight, and Agent Platform
  - Insight-to-action workflows
- Unified Data Lakes for AI
  - Petabyte-exabyte scale storage
  - Data lake-house architecture supporting both analytics and AI
- Vector and Embedding Storage
  - Storage for billions of embeddings
  - High performance indexing and metadata storage
- Storage for AI Training & Model Pipelines
  - High-throughput NVMe storage feeding GPU cluster
  - Parallel file systems and distributed storage
- Intelligent Tiered Data Lifecycle Storage
  - Auto-tiering for hot, warm and cold data



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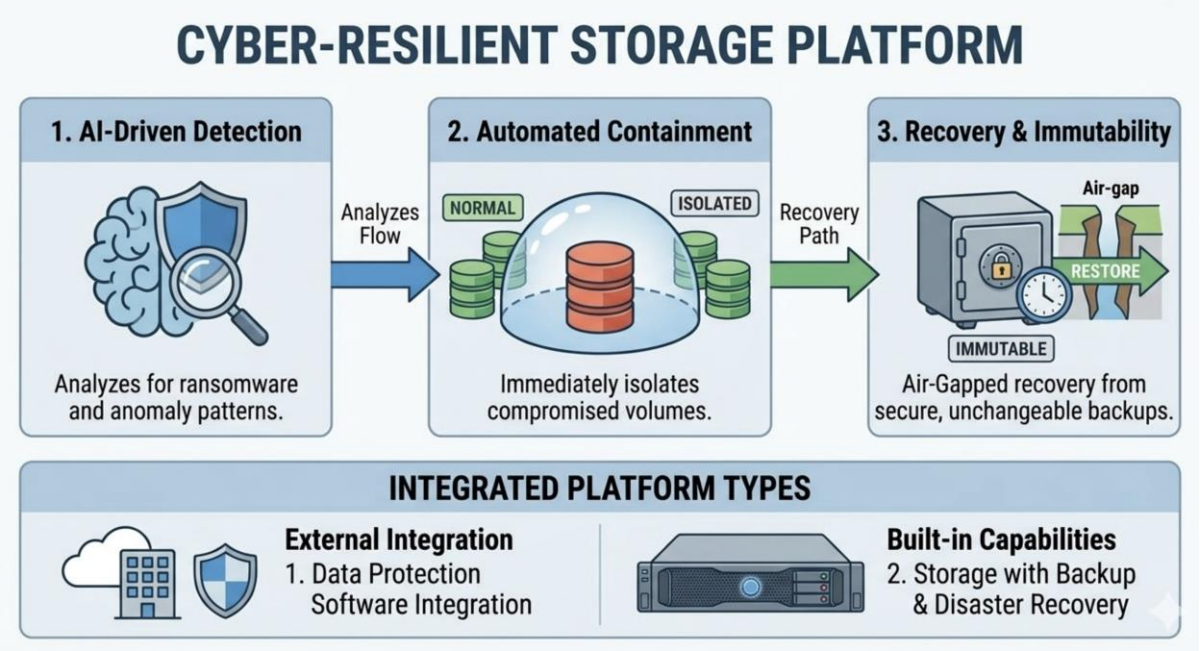
Storage #5: Cybersecure Storage

- Cyber-resilient storage platform
  - AI-driven ransomware detection in storage systems
  - Anomaly detection in file access patterns
  - Automated isolation of compromised data volumes
  - Immutable storage and ransomware-proof backups
  - Air-gapped fast recovery
- AI-driven data integrity monitoring
  - Abnormal data access detection
  - Early ransomware pattern detection
- Data protection platforms integrated with storage
  - Storage platforms include built-in backup, disaster recovery and cyber-resilience capabilities



Centralized vs De-Centralized Intelligence?

众行远
Go Far Go Together